

Ellipses and Hyperbolas in Real Structures

Reading vertices, axes, foci and asymptotes off ellipse and hyperbola models of real structures

- Every equation below models a real object, with the centre of the object at the origin and the units stated in the question.
 - Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: vertices at $\pm a$ along the longer axis, foci at $\pm c$ with $c^2 = a^2 - b^2$ (subtract).
 - Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$: vertices at $\pm a$, foci at $\pm c$ with $c^2 = a^2 + b^2$ (add), asymptote slopes $\pm b/a$.
 - Give every length with its unit, and keep values exact.
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1. The floor of a whispering gallery is an ellipse. With the centre of the room at the origin and both coordinates in metres, its edge satisfies $\frac{x^2}{169} + \frac{y^2}{144} = 1$.

(a) Find the length of the major axis, that is, the greatest width of the room.

Answer: _____ m

(b) Find the length of the minor axis.

Answer: _____ m

2. The gallery of problem 1 has two “listening spots”, one at each focus of the ellipse. Using the same equation $\frac{x^2}{169} + \frac{y^2}{144} = 1$, with coordinates in metres, find the distance between the two listening spots.

Answer: _____ m

3. A semi-elliptical bridge arch spans 40 feet at its base and rises 12 feet at the centre. Taking the centre of the span as the origin, the arch satisfies $\frac{x^2}{400} + \frac{y^2}{144} = 1$ with $y \geq 0$, both coordinates in feet. Find the height of the arch at a point 10 feet horizontally from the centre. Leave the answer exact.

Answer: _____ ft

4. The waist of a cooling tower is modelled by $\frac{x^2}{64} - \frac{y^2}{36} = 1$, with the narrowest ring of the tower at the origin and both coordinates in metres.

(a) Find the distance across the tower at its narrowest point, that is, between the two vertices.

Answer: _____ m

(b) Find the distance from the centre to each focus.

Answer: _____ m

5. The same cooling tower, $\frac{x^2}{64} - \frac{y^2}{36} = 1$ with coordinates in metres, has two asymptotes.

(a) Find the slope of the asymptote that rises from left to right.

Answer: _____

(b) Explain what those asymptotes describe about the shape of the tower wall high above the waist.

Answer: _____

6. A planetarium has an elliptical dome whose horizontal cross-section at floor level satisfies $\frac{x^2}{100} + \frac{y^2}{36} = 1$, with coordinates in metres.

(a) Find the distance from the centre to each focus.

Answer: _____ m

(b) A cable is run from one focus to a point on the ellipse and back to the other focus. Find the total cable length, which is the same wherever that point is chosen.

Answer: _____ m

7. The cross-section of a hyperbolic mirror is modelled by $\frac{y^2}{9} - \frac{x^2}{16} = 1$, with coordinates in centimetres and the mirror's axis vertical.

(a) Find the distance from the centre to each vertex.

Answer: _____ cm

(b) Find the distance from the centre to each focus.

Answer: _____ cm

8. A lamp throws a hyperbolic shadow line on a wall, modelled by $\frac{y^2}{25} - \frac{x^2}{144} = 1$ with coordinates in centimetres.

(a) Find the slope of the asymptote that rises from left to right.

Answer: _____

(b) Find the distance from the centre to each vertex.

Answer: _____ cm

9. An elliptical garden bed measures 30 feet along its longer axis and 18 feet along its shorter axis. Two sprinkler heads are to be placed at the foci of the ellipse. Find the distance between the two sprinkler heads.

Answer: _____ ft

10. A new whispering gallery is being designed. Its ceiling cross-section must be a semi-ellipse 40 metres wide at floor level, and the two listening spots, at the foci, must sit 32 metres apart.

(a) Find the required height of the ceiling above the centre of the floor.

Answer: _____ m

(b) Explain why moving the two listening spots further apart, with the width unchanged, would force a lower ceiling.

Answer: _____